Jakob Coughlan
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Experience

Systems Engineer II July 2023 - Present
Rees Scientific *Ewing, NJ*

- Designed and programmed an Electron/React desktop application capable of checking online distributions for firmware updates, executing secure flashes, and authenticating users via existing corporate SSO infrastructure.
- Designed custom automation and low voltage solutions utilizing Productivity1000 PLCs and in-house products to meet individual project requirements, complete with CAD drawings and submittal packages
- Using Fusion 360, created prototypes of several iterations of low-pressure molding for new probe designs
- Proto-typed revisions of 3 different existing products using Eagle, adding new functionality such as external power indicators and powered relays, transitioning from through-hole components to SMD, standardizing input voltage to 24VDC, and consolidating multiple large PCBs into a single compact design, reducing volume by 80%
- Engineered and deployed Power Automate flows to streamline routine operations, resulting in annual labor cost savings exceeding $32,000
- Authored and executed standardized test plans to evaluate the reliability of new and existing product lines, maintaining full alignment with ISO-compliant documentation and verification procedures.
- Created and distributed technical documents for assorted business projects using AutoCAD and Fusion 360
- Established, managed, and executed a comprehensive process for developing user-friendly and accurate assembly instructions for different products
- Developed 700 bills of material (BOMs) and several comprehensive naming schemas for all sold products used in Netsuite by collaborating with senior members of the company

System Administrator June 2019 – March 2022
Team Logic IT *Flemington, NJ*

- Managed IT systems such as Active Directory, Netgear switches and firewalls, and VMWare virtual machines for 30 companies
- Provided efficient end-user support for Windows, Macs and Office 365 issues
- Led successful data migration initiatives for multiple businesses, seamlessly transitioning to SharePoint

Education

Rutgers, The State University of New Jersey New Brunswick, NJ
Bachelor of Science in Information Technology and Informatics GPA:3.8
Minor in Philosophy

Projects

Online Multiplayer Drawing Game

- Engineered a collaborative multiplayer drawing game using React, implementing a grid-based canvas system where players draw individual tiles to complete a cohesive image, backed by a custom API for round-based game flow control and server-side asset storage.

LED Matrix Display with Real-Time Data and Custom Web Interface

- Developed a multi-functional display using a Raspberry Pi and dual LED matrices, showcasing the time, weather, and a news ticker utilizing free APIs. Additionally, the Raspberry Pi hosts a web interface for changing user display preferences such as brightness

TikTok Reddit Content Bot

- Using Python, this program retrieves content from several subreddits then automatically converts that content into a watchable video and uploads it to Tik Tok. This bot account has achieved 1000 followers and 10k Likes in one month

Social Media Moderation Bot

- Using machine learning, this bot filters users based on their profile pictures, username, and content output to prevent spamming. This bot has filtered 100 bad actors from interacting with honest users